Statistical Tests, Formulae, Confirmed Outputs, and Interpretation

Appendix-ready table for the NeurIPS submission. This document separates (i) core locked brain-side tests from (ii) supporting cross-system tests and (iii) values that must be consistency-checked before final manuscript wording. Formulae are written in plain mathematical notation so they can be converted directly into LaTeX.

0. Notation used throughout

Symbol	Meaning in this paper
x_i in R^d	Representation vector for sample i. For brain, d=48 ROI features in the main five-emotion dataset; reduced 11D vectors were used in the specific margin/uncertainty validation notebook. For LLMs, d is the embedding dimension.
y_i	Emotion label for sample i. Brain labels: Afraid, Calm, Delighted, Depressed, Excited.
c_k	Centroid/prototype of emotion class k.
d_correct	Distance from sample x_i to the centroid of its labelled/correct class.
d_nearest_other	Distance from sample x_i to the closest centroid whose label is not y_i.
margin	d_correct - d_nearest_other. Negative = closer to correct class; near zero = class competition; positive = closer to another class.
uncertainty	1 - P(correct), where P(correct) is the cross-validated classifier probability assigned to the true label.

1. Core geometry and ambiguity statistics

Test / metric	Formula used	Confirmed output	Meaning for the paper
Class centroid	c_k = (1/N_k) * sum_{i: y_i=k} x_i	No single numeric output; computed for each emotion class.	Defines the prototype/average representation for each emotion. Centroids are statistical prototypes, not pure biological states.
Euclidean distance to a centroid	d(x_i,c_k) = ||x_i - c_k||_2 = sqrt(sum_j (x_ij - c_kj)^2)	Used to compute d_correct and all pairwise centroid distances/RDMs.	Measures representational dissimilarity in feature space. Important: raw distance alone was not the main ambiguity mechanism.
Raw distance vs uncertainty - Spearman correlation	rho = Pearson(rank(d_correct), rank(uncertainty))	rho = 0.1509, p = 3.2900e-02	Statistically non-zero but weak. This shows the naive idea "far from centroid = ambiguous" is not sufficient.
Raw distance vs uncertainty - Pearson correlation	r = sum((d_i-mean(d))(u_i-mean(u))) / sqrt(sum(d_i-mean(d))^2 * sum(u_i-mean(u))^2)	r = 0.1471, p = 3.7628e-02	Weak linear relationship. Absolute distance contains little practical information about uncertainty.
Raw distance AUC for high uncertainty	AUC = P(score(x_high_uncertainty) > score(x_low_uncertainty))	AUC = 0.5627	Near chance (0.5). Distance alone is a poor separator of high vs low uncertainty samples.
Centroid margin	m_i = d(x_i,c_{y_i}) - min_{k != y_i} d(x_i,c_k)	Computed for all samples; used as primary ambiguity metric.	Measures class competition. Ambiguity is relational: a point is uncertain when it is similarly close to its own class and a competing class.
Margin vs uncertainty - Spearman correlation	rho = Pearson(rank(margin), rank(uncertainty))	rho = 0.6108, p = 7.78e-22	Strong monotonic relationship. This is the core evidence that uncertainty is governed by relative centroid competition.
Margin vs uncertainty - Pearson correlation	r = corr(margin, uncertainty)	r = 0.6351	Strong linear association between margin and uncertainty.
Margin AUC for high uncertainty	AUC = P(margin_high_uncertainty > margin_low_uncertainty)	AUC = 0.8124	Strong predictive power. Margin separates high-uncertainty from low-uncertainty samples far better than raw distance.
Feature-shuffle control for margin	Shuffle each feature column independently, recompute margin, then test rho(margin_shuffled, uncertainty)	rho = 0.0287, p = 6.87e-01	The margin-uncertainty relationship collapses under randomisation. This supports the claim that the effect depends on real multivariate geometry, not random feature distributions.

Paper-ready interpretation: Absolute distance is a weak predictor of uncertainty, whereas centroid margin is strong. Therefore, ambiguity should be framed as a competitive/relational property of the representation space, not as simple dispersion away from a centroid.

2. Brain decoding and statistical validation

Test / metric	Formula used	Confirmed output	Meaning for the paper
LOSO accuracy	Accuracy = (1/n) * sum_i 1(y_hat_i = y_i), with one subject held out per fold	Accuracy = 0.5600; chance = 0.2000	The 48D brain ROI space contains emotion-decodable information above five-class chance.
LOSO balanced accuracy	BalancedAcc = (1/K) * sum_k TP_k / (TP_k + FN_k)	Balanced accuracy = 0.5600	Performance is not driven by one emotion class. Each class contributes equally to the average.
Bootstrap CI for accuracy	Resample predictions with replacement; CI = percentile_{2.5,97.5}(metric_bootstrap)	Accuracy 95% CI = [0.4900, 0.6300]	Provides uncertainty around decoding performance. The interval remains well above 0.20 chance.
Bootstrap CI for balanced accuracy	Same percentile bootstrap applied to balanced accuracy	Balanced accuracy 95% CI = [0.4906, 0.6280]	Shows robustness of class-balanced decoding estimate.
LOSO permutation test	Shuffle labels, recompute LOSO accuracy; p = (# null >= observed + correction)/(N_perm + correction)	Direct session output: null mean = 0.197825, null SD = 0.0334364, p-value reported as 0.0	Observed decoding is far above the shuffled-label null. In manuscript, report as p < 1/N_perm or use add-one correction rather than literal p=0.0.
Subject-level accuracy summary	Per-subject accuracy = correct predictions / 5 emotion samples	Mean = 0.5600; SD = 0.240512; min = 0.20; median = 0.60; max = 1.00	Shows variability across subjects. Useful for limitations/appendix, not the main claim.

3. Valence-arousal and global geometry statistics

Test / metric	Formula used	Confirmed output	Meaning for the paper
PCA explained variance on brain centroids	ExplainedVar_j = lambda_j / sum_l lambda_l	Brain 5-emotion notebook: PC1 = 0.46421237; PC2 = 0.32432139	The first two linear axes capture approximately 78.85% of centroid variation. This supports a low-dimensional visual summary, not a replacement for high-dimensional analysis.
Brain PC1 vs valence	Pearson r between PC1 centroid coordinates and assigned valence scores	r = 0.255, p = 0.6789	No meaningful evidence that the dominant brain axis is valence-aligned in this five-centroid analysis.
Brain PC1 vs arousal	Pearson r between PC1 centroid coordinates and assigned arousal scores	r = 0.965, p = 0.0078	Strong evidence that the dominant brain centroid axis aligns with arousal. Caveat: only five centroids.
Brain PC2 vs valence	Pearson r between PC2 coordinates and valence scores	r = -0.064, p = 0.9185	No meaningful PC2-valence alignment.
Brain PC2 vs arousal	Pearson r between PC2 coordinates and arousal scores	r = 0.039, p = 0.9504	No meaningful PC2-arousal alignment.
Cross-system VA: Qwen-768	Correlation of 2D embedding geometry with valence/arousal ideal map; permutation test for alignment	valence r = 0.9822; arousal r = 0.8249; Procrustes disparity = 0.1823; permutation p = 0.013	Qwen geometry is strongly valence-aligned and close to the ideal VA map.
Cross-system VA: MPNet-Balanced	Same as above	valence r = 0.9655; arousal r = 0.7326; Procrustes disparity = 0.1949; permutation p = 0.009; bootstrap CI for valence r = [0.9227, 0.99999]	MPNet also shows strong valence-first geometry.
Cross-system VA: Brain-fMRI from alignment_metrics.json	Same cross-system VA alignment implementation	valence r = 0.3141; arousal r = 0.9565; Procrustes disparity = 0.8033; permutation p = 0.033; bootstrap CI = [0.2080, 1.0]	Brain geometry is arousal-aligned but less close to the ideal 2D VA map. Do not mix this with the separate notebook p=0.0078 without explaining implementation differences.

4. RDM, centroid, density, and retention statistics

Test / metric	Formula used	Confirmed output	Meaning for the paper
Representational Dissimilarity Matrix (RDM)	R_{ab} = d(c_a, c_b)	Brain centroid distances include: Afraid-Excited = 0.9634; Calm-Excited = 1.5073; Calm-Delighted = 1.3902; Afraid-Depressed = 1.2997; see appendix table for all pairs.	RDM captures global category-level geometry. It is not the same as sample-level ambiguity.
Brain vs Qwen centroid relational Pearson correlation	r = corr(vectorized upper triangle of Brain RDM, vectorized upper triangle of Qwen RDM)	centroid_relational_report.html: Pearson r = -0.8756; bootstrap CI = [-0.9967, 0.6994]	Suggests inverted global relational geometry, but the wide CI means this should be supporting, not central. Verify consistency with any JSON outputs before final paper.

Test / metric	Formula used	Confirmed output	Meaning for the paper
Manhattan robustness for centroid relation	Same RDM comparison using L1/Manhattan distances	r = -0.5476	Shows the negative trend is not purely Euclidean-distance dependent, but still should be framed cautiously.
Density geometry - Wasserstein distance	WD(P,Q) = integral |F_P(t)-F_Q(t)| dt for 1D distributions	Brain vs Qwen WD = 0.0904; bootstrap 95% CI = [0.0781, 0.1087]; brain-brain noise ceiling = 0.0326	Indicates density distributions are relatively similar, though not identical. Use as supporting geometry, not core mechanism.
Density permutation p-value	Permutation p = proportion of shuffled/null density statistics as extreme as observed	p = 0.4600	Not significant. Interpret as: density decay alone is not sufficient evidence for the ambiguity mechanism. Do not write that p=0.46 proves a law.
Iterative retention AUC / signal volume	AUC over accuracy-retention curve after sequential removal of discriminative directions	MPNet AUC = 0.3305, D50 = 17 dims; Qwen AUC = 0.2600, D50 = 12 dims; Brain AUC = 0.3021, D50 = 4 dims	Shows different representational regimes. Be careful: Brain D50=4 suggests early signal loss despite moderate AUC. Avoid overclaiming brain redundancy.
Cliff slope	Approximate rate of performance decline per removed direction	MPNet = 0.0210; Qwen = 0.0311; Brain = 0.0225	Supporting description of decay profile; not central to the margin mechanism.

5. All brain centroid-pair RDM distances recorded

Pairwise brain centroid distance	Value	Interpretation
Afraid - Calm	1.1097	Moderate separation.
Afraid - Delighted	1.1826	Moderate separation.
Afraid - Depressed	1.2997	Larger separation.
Afraid - Excited	0.9634	Closest listed Afraid relation; suggests similar high-arousal structure.
Calm - Delighted	1.3902	Large separation.
Calm - Depressed	1.0423	Moderate/low separation.
Calm - Excited	1.5073	Largest listed separation.
Delighted - Depressed	1.0419	Moderate/low separation.
Delighted - Excited	1.1244	Moderate separation.
Depressed - Excited	1.4554	Large separation.

6. Manuscript reporting rules

Rule	Reason
Never report p = 0.0 literally.	Finite permutation tests cannot prove zero probability. Write p < 0.001 or p < 1/(N_perm+1), depending on the rerun.
Use margin, not raw distance, as the formal ambiguity metric.	Raw distance is weak; margin is the validated predictor of uncertainty.
Keep density p=0.46 as a negative/supporting result.	A non-significant p-value cannot confirm a density law. It supports the conclusion that global density is not the core ambiguity mechanism.
Do not mix VA outputs from different implementations without labels.	Brain PC1-arousal p=0.0078 and alignment_metrics p=0.033 likely come from related but not identical analyses. State which one each table uses.
Report uncertainty intervals where available.	NeurIPS checklist asks for error bars, CIs, or significance tests supporting main claims.
Tie every formula to an interpretation.	Reviewers should see not only what was computed, but why it supports the claim.

Final appendix wording: The central statistical result is that centroid margin strongly predicts classifier uncertainty (Spearman rho = 0.6108, $p = 7.78\mathrm{e}{-22}$; AUC = 0.8124), whereas absolute distance to the correct centroid is weak (rho = 0.1509; AUC = 0.5627). The effect disappears after feature shuffling (rho = 0.0287, $p = 0.687$), supporting the interpretation that ambiguity reflects real multivariate class competition rather than random dispersion.